

FUJITSU

CMOS 1,048,576 BIT UV ERASABLE READ ONLY MEMORY (EPROM)

MBM27C1001-15
MBM27C1001-20
MBM27C1001-25

April 1988
Edition 2.0

CMOS 1,048,576 BIT UV ERASABLE READ ONLY MEMORY (EPROM)

The Fujitsu MBM27C1001 EPROM is a high speed read-only static memory that is UV-erasable and reprogrammable. The device contains 1,048,576 programmable or reprogrammable bits organized in a 131,072-byte/8-bit format. The MBM27C1001 is housed in a 32-pin DIP and 36-pad LCC with a transparent lid; when the lid is properly exposed to an ultraviolet light source, a previously programmed bit pattern is erased in approximately 15-to-21 minutes. A new bit pattern can then be written into memory.

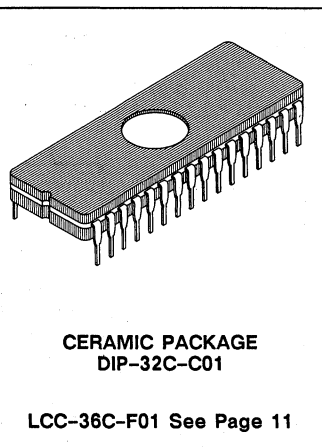
The MBM27C1001 EPROM is fabricated using CMOS double poly-silicon gate technology with stacked single-transistor gate cells. The MBM27C1001 is an excellent choice for system development work and in other applications where program changes are frequently necessary. Once programmed, the device requires only a single +5V power supply; the current requirements are exceptionally low in both the active and standby modes of operation.

- 131,072-byte/8-bit organization with on-chip decoding
- Single-byte or four-byte programming capability with Quick Pro™ algorithm
- Static operation (no clocks required)
- Upward compatible with 256K/512K EPROMS
- Fast access time:
MBM27C1001-15 = 150 ns (max)
MBM27C1001-20 = 200 ns (max)
MBM27C1001-25 = 250 ns (max)
- Easy and simple memory expansion via @pin
- Three-state output for wired-OR capability
- TTL-compatible inputs/outputs
- Single =5V (+10%) power supply with low current drain:
Active operation = 30 mA (max) for 200ns/250ns
40 mA (max) for 150ns
Standby operation = 0.1 mA (max)
- Programming voltage: +12.5V
- JEDEC-approved pin assignments
- 32-pin CERDIP

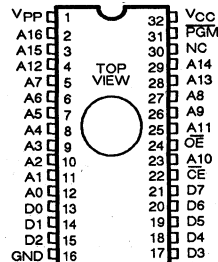
ABSOLUTE MAXIMUM RATINGS (see NOTE)

Rating	Symbol	Value	Unit
Supply Voltage with respect to ground	V _{CC}	-0.6 to +7.0	V
Programming Voltage with respect to ground	V _{PP}	-0.6 to +14.0	V
Input/Output Voltage (except for A ₀ with respect to ground)	V _{IN1}	-0.6 to V _{CC} + 0.3	V
Programming Voltage with respect to ground	V _{IN2}	-0.6 to +13.5	V
Temperature under Bias	T _{BIAS}	-25 to +85	°C
Storage Temperature Range	T _{STG}	-65 to +125	°C

NOTE: Permanent device damage may occur if the above Absolute Maximum Ratings are exceeded. Functional operation should be restricted to the conditions as detailed in the operational sections of this data sheet. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.



PIN ASSIGNMENT

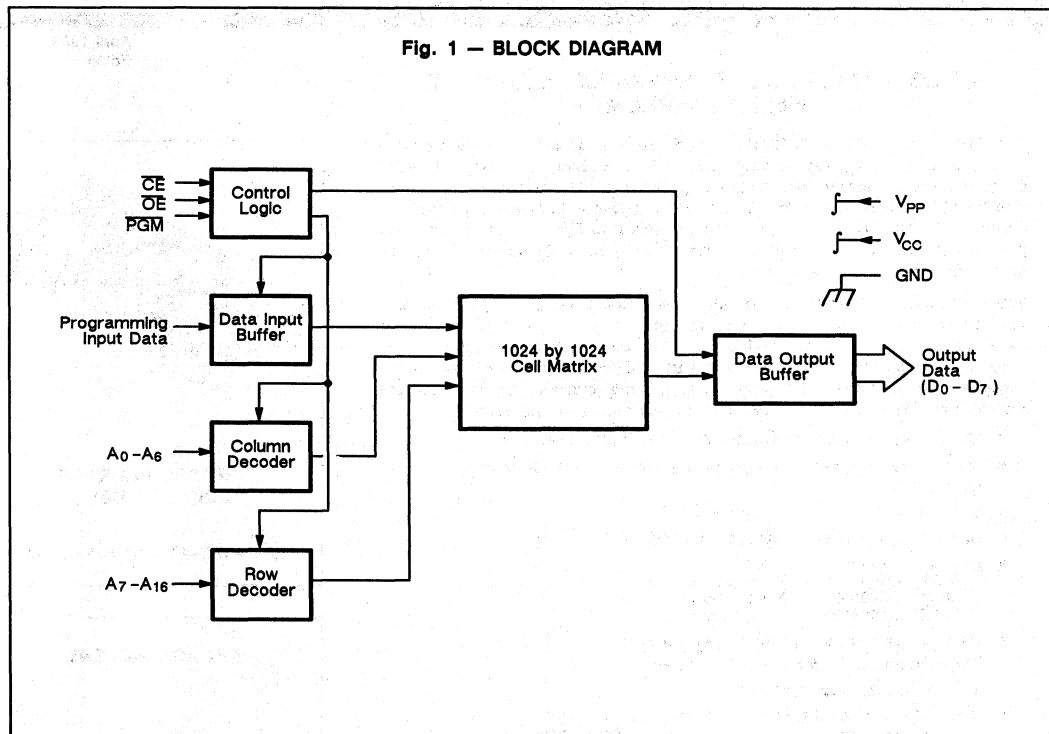


This device contains circuitry to protect the inputs against damage due to high static voltages or electric fields. However, it is advised that normal precautions be taken to avoid application of any voltage higher than maximum rated voltages to this high impedance circuit.

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Fig. 1 — BLOCK DIAGRAM



CAPACITANCE (T_A = 25 °C, f = 1MHz)

Parameter	Symbol	Values			Unit
		Min	Typ	Max	
Input Capacitance (V _{IN} = 0V)	C _{IN}			12	pF
Output Capacitance (V _{OUT} = 0V)	C _{OUT}			12	pF

PIN DESCRIPTION

Symbol	Pin No. *	Function
V_{PP}	1	+12.5V programming voltage.
$A_0 - A_{16}$	2-12, 23 25-29	Address lines.
$O_0 - O_7$	13-15, 17-21	Three-state output data lines.
GND	16	Circuit ground.
$\overline{CE}$	22	When active Low, the device is enabled for data read.
$\overline{OE}$	24	Output enable. When $\overline{OE}$ and $\overline{CE}$ are active low and the $\overline{PGM}$ strobe is active High; all output lines ($D_0 - D_7$) are enabled.
NC	30	No connection.
$\overline{PGM}$	31	When active Low, programming data from the input buffer is written into a specified address of memory.
V_{CC}	32	+5V power supply

* This numbers are applied to DIP package.

FUNCTIONS AND PIN CONNECTIONS

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OPERATING MODE	$A_0 - A_8$	A_9	$A_{10} - A_{16}$	Data	$\overline{CE}$	$\overline{OE}$	$\overline{PGM}$	V_{CC}	V_{PP}	GND
Standby	X	X	X	HI-Z	V_{IH}	X	X	5V	5V	0V
Read	A_{IN}	A_{IN}	A_{IN}	D_{OUT}	V_{IL}	V_{IL}	V_{IH}	5V	5V	0V
Output Disable	A_{IN}	A_{IN}	A_{IN}	HI-Z	V_{IL}	V_{IH} X V_{IL}	X V_{IL}	5V	5V	0V
Electronic Signature	Note 1	12V	X	CODE	V_{IL}	V_{IL}	V_{IH}	5V	5V	0V
Single Byte Program	A_{IN}	A_{IN}	A_{IN}	D_{IN}	V_{IL}	V_{IH}	V_{IL}	6V	12.5V	0V
Single Byte Verify	A_{IN}	A_{IN}	A_{IN}	D_{OUT}	V_{IL}	V_{IL}	V_{IH}	6V	12.5V	0V
Single Byte Program Inhibit	A_{IN}	A_{IN}	A_{IN}	HI-Z	V_{IL}	V_{IH}	V_{IH}	6V	12.5V	0V
Four Byte Data Input	Note 2	A_{IN}	A_{IN}	D_{IN}	V_{IH}	V_{IH}	V_{IH}	6V	12.5V	0V
Four Byte Program	X	A_{IN}	A_{IN}	HI-Z	V_{IH}	V_{IL}	V_{IL}	6V	12.5V	0V
Four Byte Verify	Note 2	A_{IN}	A_{IN}	D_{OUT}	V_{IL}	V_{IL}	V_{IH}	6V	12.5V	0V
Four Byte Program Inhibit	A_{IN}	A_{IN}	A_{IN}	HI-Z	V_{IL}	V_{IH}	V_{IH}	6V	12.5V	0V

Legend:

X = Don't care

Notes:

1. A_0 is toggling address. A_1 is V_{IL} .
2. A_0 and A_1 can be either V_{IL} or V_{IH}

RECOMMENDED OPERATING CONDITIONS

(Referenced to GND)

Parameter	Symbol	Min	Typ	Max	Unit
Supply Voltage	V_{CC}	4.5	5.0	5.5	V
Supply Voltage	V_{PP}	$V_{CC} - 0.6$	V_{CC}	$V_{CC} + 0.6$	V
Input High Level	V_{IH}	2.0		$V_{CC} + 0.3$	V
Input Low Level	V_{IL}	-0.1		0.8	V
Supply Voltage	GND		0		V
Operating Temperature	T_A	0		70	°C

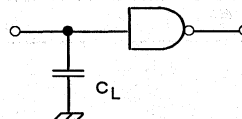
DC CHARACTERISTICS

(Recommended operating conditions unless otherwise noted)

Parameter	Symbol	Conditions	Values			Unit
			Min	Typ	Max	
Input Leakage Current	I_{LI}	$V_{IN} = V_{CC} = 5.5V$			10	μA
Output Leakage Current	I_{LO}	$V_{IN} = V_{CC} = 5.5V$			10	μA
V_{CC} Standby Current	I_{SB1}	$\overline{CE} = V_{IH}$			1	mA
V_{CC} Standby Current	I_{SB2}	$\overline{CE} = V_{CC} \pm 0.3V$		1	100	μA
V_{CC} Active Current	I_{CC1}	$\overline{CE} = V_{IL}, I_{OUT} = 0mA$			30	mA
V_{CC} Operation Current	I_{CC2}	$\overline{CE} = V_{IL}, f = \min, I_{OUT} = 0mA$			40	mA
					30	
V_{PP} Supply Current	I_{PP}	$V_{PP} = V_{CC} \pm 0.6V$		1	100	μA
Output Low Level	V_{OL}	$I_{OL} = 2.1mA$			0.45	V
Output High Level	V_{OH1}	$I_{OH} = -400 \mu A$	2.4			V
Output High Level	V_{OH2}	$I_{OH} = -100 \mu A$	$V_{CC} - 0.7$			V

Fig. 2 — AC TEST CONDITIONS (INCLUDING PROGRAMMING)

Input pulse levels: 0.45V to 2.4V (0.3V to 2.8V programming)
 Input Rise/Fall Times: $\leq 20ns$
 Input Reference Levels: 0.8V to 2.0V (0.6V to 2.4V programming)
 Output Reference Levels: 0.8V to 2.0V
 Output Load: 1 TTL gate and $C_L = 100pF$

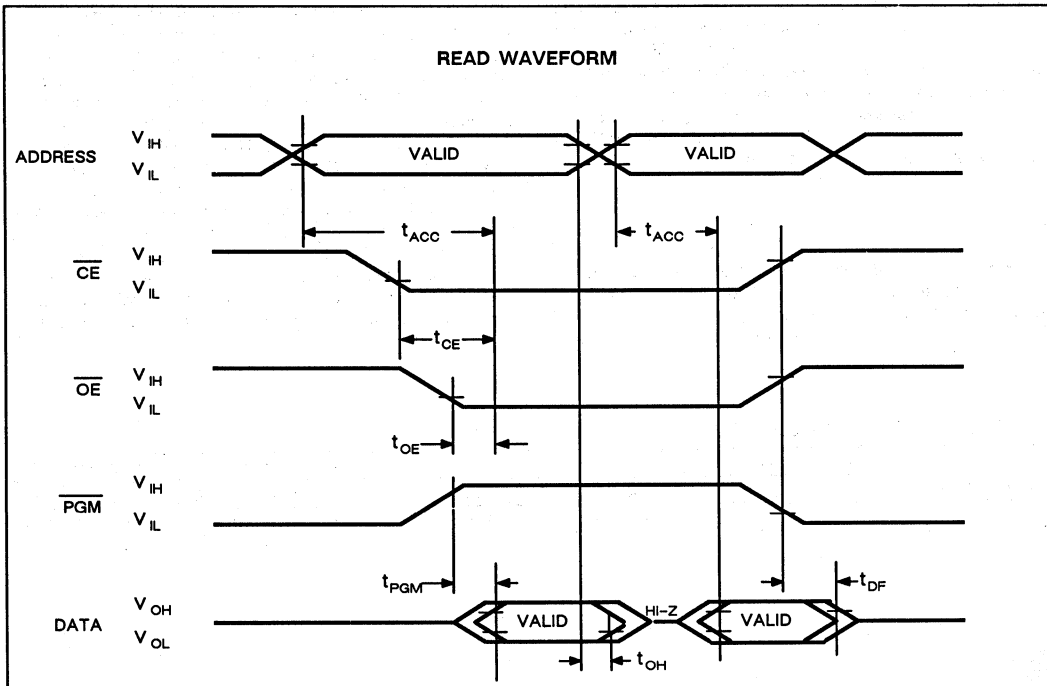


AC CHARACTERISTICS

(Recommended operating conditions unless otherwise noted)

Parameter	Symbol	MBM27C1001-15 Values		MBM27C1001-20 Values		MBM27C1001-25 Values		Unit
		Min	Max	Min	Max	Min	Max	
Address Access Time	t_{ACC}		150		200		250	ns
$\overline{CE}$ to Output Delay Time	t_{CE}		150		200		250	ns
$\overline{OE}$ to Output Delay Time	t_{OE}		70	0	70	0	100	ns
$\overline{PGM}$ to Output Delay Time	t_{PGM}		70	0	70	0	100	ns
$\overline{CE}$ or $\overline{OE}$ to Output Float Delay (Note)	t_{DF}		60		60		60	ns
Address to Output Hold Time	t_{OH}	0		0		0		ns

NOTE: Output Float is defined as the point where data is no longer driven.





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PROGRAMMING / ERASING INFORMATION

PROGRAMMING

Single-Byte Programming. When +12.5V(± 0.3) volts is applied to V_{PP} , +6(± 0.25) volts is applied to V_{CC} , $\overline{CE}$ and $\overline{PGM} = V_{IH}$, and $\overline{OE} = V_{IH}$, the programming mode is initiated. Next, the proper address is input and the data pattern is applied to the input buffer (Figure 1). When both address and data are stable, a 0.5-millisecond negative pulse is applied to the $\overline{PGM}$ pin. Upon verification of written data an over pulse (three times the initial pulse width times the number of pulses used to accomplish a write) should be applied to complete the programming of one byte. Refer to the PROGRAMMING FLOWCHART that follows for step-by-step programming procedures.

Four-Byte Programming. When compared to single-byte programming, the four-byte programming method reduces the programming time by about 75% one quarter. Voltages applied to V_{PP} and V_{CC} are the same as those for single-byte programming; however, some logic levels differ—refer to "Four Byte Programming" in the Truth Table. In conjunction with the $\overline{OE}$ pin, address pins A0 and A1 are used to latch four bytes of data. When both address and data are stable, a 0.5 millisecond negative pulse is applied to the $\overline{PGM}$ pin. Upon verification of written data an over pulse (three times the initial pulse width times the number of pulses used to accomplish a write,) should be applied to complete the programming of four bytes. Refer to the PROGRAMMING FLOWCHART for step-by-step programming procedures.

Caution

The width of one programming pulse must not exceed 40-millisecond; thus, a continuous TTL low-level voltage should not be applied to the $\overline{PGM}$ pin. Also, a 0.1-microfarad capacitor must be connected between V_{PP} and ground to

prevent excessive voltage transients. Neglecting either of these precautions may cause device failure.

Electronic Signature/Programming Algorithm. When the MBM27C1001 is shipped from the factory, all memory cells (1,048,576 bits) are set to the High state (logic 1). During the programming procedure, affected bit cells are set to the Low (logic 0) state.

The MBM27C1001 is programmed with a fast programming algorithm designed by Fujitsu called Quick Pro™. Manufacturer and device codes are electronically stored in each device; these codes can be read at the output port (D0-D7) for the purpose of matching the device with the Quick Pro™ algorithm. The Electronic Signature Code List is shown preceding the ELECTRICAL CHARACTERISTICS.

ERASING

In order to clear all memory cells of programmed contents, the MBM27C1001 must be exposed to an ultraviolet light source. To completely erase the memory (restore all cells to a logic 1 state), a dosage of 15Wsec/cm² is required. The required exposure can be obtained by using a UV-lamp with a wavelength of 2537 Angstroms and with an intensity of 12mW/cm². Remove all filters from the lamp and clean the transparent lid of the MBM 27C1001 with a non-abrasive cleaner. Hold the MBM 27C1001 approximately one inch from the light source for 15-to-21 minutes. (Note. The MBM 27C1001 and other similar devices can be erased by light sources with longer wavelengths; however, the erasing time is much greater. Nonetheless, exposure to fluorescents or sunlight will severely degrade and eventually erase the memory. When used in a lighted environment, it is recommended that the transparent window be covered with an opaque label.)

ELECTRONIC SIGNATURE CODE LIST

Definition	AO	A1 TO A6	A9	A7 to A16	D0	D1	D2	D3	D4	D5	D6	D7	HEX
Manufacture	VIL	VIL	12(± 0.5)V	Don't Care	0	0	1	0	0	0	0	0	#04
Device	VIH	VIL	12(± 0.5)V	Don't Care	0	1	1	0	0	1	1	1	#E6

The timing diagram illustrates the relationship between several signals during a memory access cycle. The signals shown are:

- V_{PP}**: Program Voltage, which is high during programming and high impedance during read.
- V_{CC}**: Supply Voltage, which is high during both read and program.
- ADDRESS**: Consists of **V_{IH}** (high address) and **V_{IL}** (low address) lines.
- DATA**: Consists of **V_{IH}/V_{OH}** (data input/output high) and **V_{IL}/V_{OL}** (data input/output low) lines.
- OE** (Output Enable): Active low signal that enables the data output.
- CE** (Chip Enable): Active low signal that enables the device.
- PGM** (Program Enable): Active low signal used to initiate programming.

Key timing parameters and events marked on the diagram include:

- t_{VPS}**: Address setup time before V_{PP} is applied.
- t_{AH}**: Address hold time after V_{PP} is removed.
- t_{AS}**: Address setup time before data input becomes valid.
- t_{DS}**: Data setup time before OE is asserted.
- t_{OH}**: Data output hold time after OE is deasserted.
- t_{DS}**: Data setup time before CE is asserted.
- t_{OE}**: OE setup time before data output becomes valid.
- t_{DFV}**: Data output delay from OE assertion to data becoming valid.
- t_{OE}**: OE hold time after data output becomes invalid.
- t_{CE}**: CE setup time before data output becomes valid.
- t_{CE}**: CE hold time after data output becomes invalid.
- t_{OPW}**: Output programming time from PGM assertion to data output becoming valid.
- t_{PW}**: Programming pulse width.
- t_{DH}**: Data hold time after PGM is deasserted.

The diagram shows three distinct phases: a read cycle (DATA IN VALID), a data output phase (DATA OUT), and another read cycle (DATA IN VALID). The PGM signal is only active during the programming phase.

The timing diagram illustrates the electrical characteristics and timing parameters of the 74VHC04N hex inverter. The signals shown are:

- V_{PP} : Power supply voltage.
- V_{CC} : Supply voltage.
- $A_2 - A_{15}$: Inputs A_2 through A_{15} .
- A_1 : Input A_1 .
- A_0 : Input A_0 .
- $D_0 - D_7$: Data bus signals.
- $\overline{OE}$: Output Enable signal.
- $\overline{CE}$: Chip Enable signal.
- $\overline{PGM}$: Program signal.

The diagram shows the relationship between these signals and the timing parameters:

- t_{VPS} : Setup time for V_{CC} .
- t_{ACV} : Access time for V_{CC} .
- t_{HAV} : Hold time for V_{CC} .
- t_{AS} : Setup time for A_0 .
- t_{AH} : Hold time for A_0 .
- t_{DS} : Delay time for $D_0 - D_7$.
- t_{DH} : Delay time for $D_0 - D_7$.
- t_{OE} : Output Enable delay time.
- t_{HBP} : Hold time for $\overline{CE}$.
- t_{HAP} : Hold time for $\overline{CE}$.
- t_{PW} (t_{OPW}): Pulse width for $\overline{PGM}$.
- t_{DFV} : Delay time for $D_0 - D_7$.



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DC CHARACTERISTICS (Programming Mode)

($T_A = 25\text{ }^{\circ}\text{C} \pm 5^{\circ}\text{C}$, $V_{CC}^1 = 6\text{V} \pm 0.25\text{V}$, $V_{PP}^2 = 12.5\text{V} \pm 0.3\text{V}$)

Parameter	Symbol	Conditions	Values			Unit
			Min	Typ	Max	
Input Leakage Current	I_{LI}	$V_{IN} = 6.25\text{V}/0\text{V}$			10	μA
Input High Level	V_{IH}		2.4		$V_{CC} + 0.3$	V
Input Low Level	V_{IL}		-0.1		0.6	V
V_{CC} Supply Current	I_{CC}				30	mA
V_{PP} Supply Current	I_{PP1}	$\overline{CE} = \overline{PGM} = V_{IL}$; $\overline{OE} = V_{IH}$			30	mA
V_{PP} Supply Current	I_{PP2}	$\overline{CE} = V_{IH}$; $\overline{OE} = \overline{PGM} = V_{IL}$			100	mA
V_{PP} Supply Current	I_{PP3}	$\overline{PGM} = V_{IH}$			5	mA
Output Low Level	V_{OL}	$I_{OL} = 2.1\text{mA}$			0.45	V
Output High Level	V_{OH}	$I_{OH} = -400\text{ }\mu\text{A}$	2.4			V

NOTE *1 V_{CC} must be applied either coincidentally or before V_{PP} and removed either coincidentally or after V_{PP} .

*2 V_{PP} must not be greater than 13.5 volts including overshoot. Permanent device damage may occur if the device is taken out or put into socket remaining $V_{PP} = 12.5$ volts. Also, during $\overline{CE} = V_{IL}$, $\overline{OE} = V_{IH}$, V_{PP} must not be switched from V_{CC} to V_{PP} volts or vice versa.

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AC CHARACTERISTICS (Single Byte Programming)

Parameter	Symbol	Values			Unit
		Min	Typ	Max	
V_{PP} Setup Time	t_{VPS}	2			μS
Address Setup Time	t_{AS}	2			μS
Data Setup Time	t_{DS}	2			μS
$\overline{CE}$ Setup Time	t_{CES}	2			μS
$\overline{OE}$ Setup time	t_{OES}	2			μS
Address Hold Time	t_{AH}	2			μS
Data Hold Time	t_{DH}	2			μS
$\overline{OE}$ to Output Valid	t_{OEV}			500	ns
$\overline{OE}$ to Output Float	t_{DFV}			150	ns
Programming Pulse Width	t_{PW}	0.475	0.50	0.525	ms
Over Programming Pulse Number	N	1		25	times
Over Programming Pulse Width (Note)	t_{OPW}	1.4	1.5	39.4	ms

NOTE: $t_{OPW} = 1.5 \times N\text{ms} \pm 5\%$

AC CHARACTERISTICS
(Four Byte Programming)

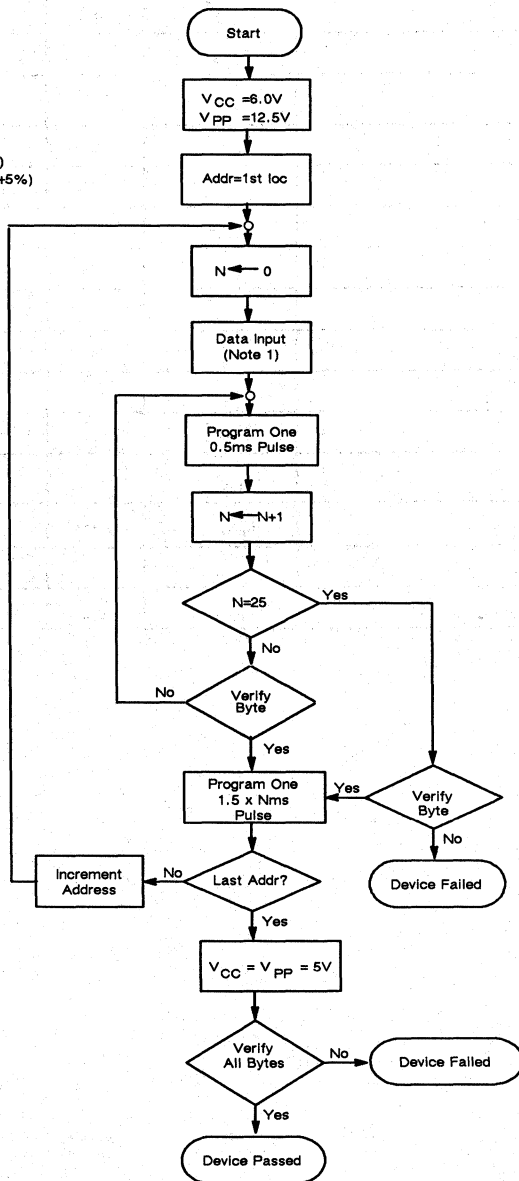
Parameter	Symbol	Values			Unit
		Min	Typ	Max	
V _{PP} Setup Time	t _{VPS}	2			μs
Address Setup Time	t _{AS}	2			μs
Data Setup Time	t _{DS}	2			μs
Address Hold Time	t _{AH}	2			μs
Data Hold Time	t _{DH}	2			μs
OE High Hold Time	t _{OEH}	2			μs
Hold Time Before Programming	t _{HBP}	2			μs
Hold Time After Program	t _{HAP}	2			μs
Address Access Time at Verify	t _{ACV}			500	ns
CE to Output Float at Verify	t _{DFV}			150	ns
Hold Time After Verify	t _{HAV}	0			μs
Programming Pulse Width	t _{PW}	0.475	0.50	0.525	ms
Over Programming Pulse Number	N	1		25	times
Over Programming Pulse Width (Note)	t _{OPW}	1.4	1.5	39.4	ms

NOTE: t_{OPW} = 1.5 × Nms ± 5%

PROGRAMMING / ERASING INFORMATION (Cont'd)

PROGRAMMING FLOWCHART FOR QUICK PRO™

Notes:
1. 1-byte or 4-bytes
2. Conditions:
 $V_{CC} = 6V(\pm 0.25)$
 $V_{PP} = 12.5(\pm 0.3)V$
 $t_{PW} = 0.5ms (+5\%)$
 $t_{OPW} = 1.5 \times Nms (+5\%)$

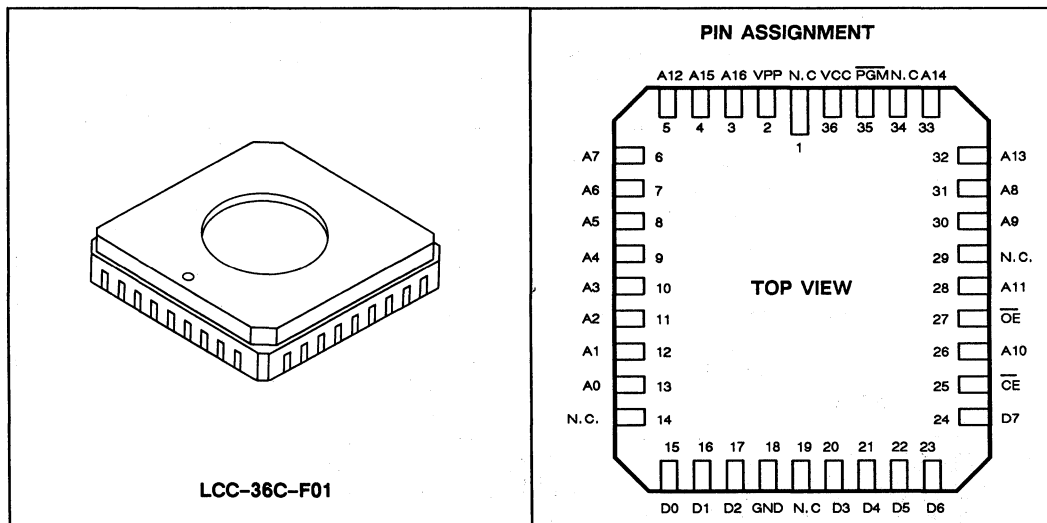


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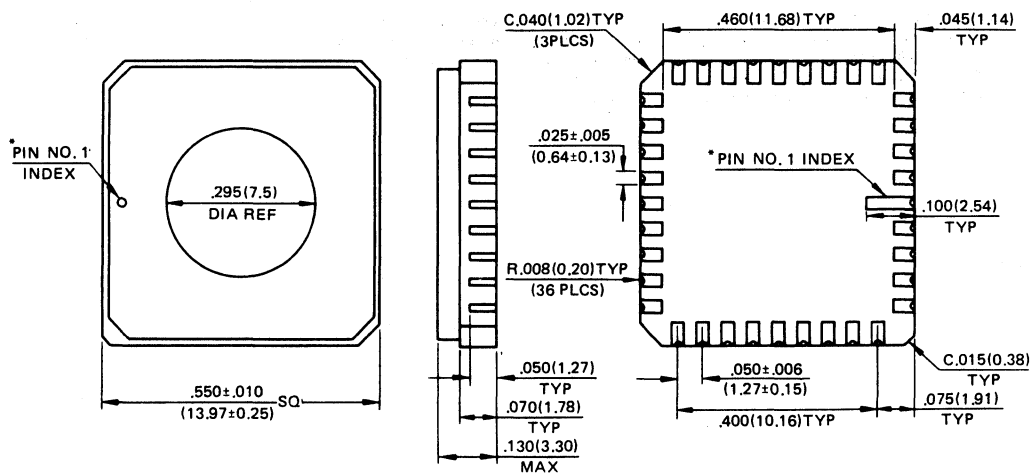
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PACKAGE DIMENSIONS



32-LEAD CERAMIC (CERDIP) DUAL IN-LINE PACKAGE (Case No.: LCC-36C-F01)



* Shape of PIN NO. 1 INDEX: Subject to change without notice.

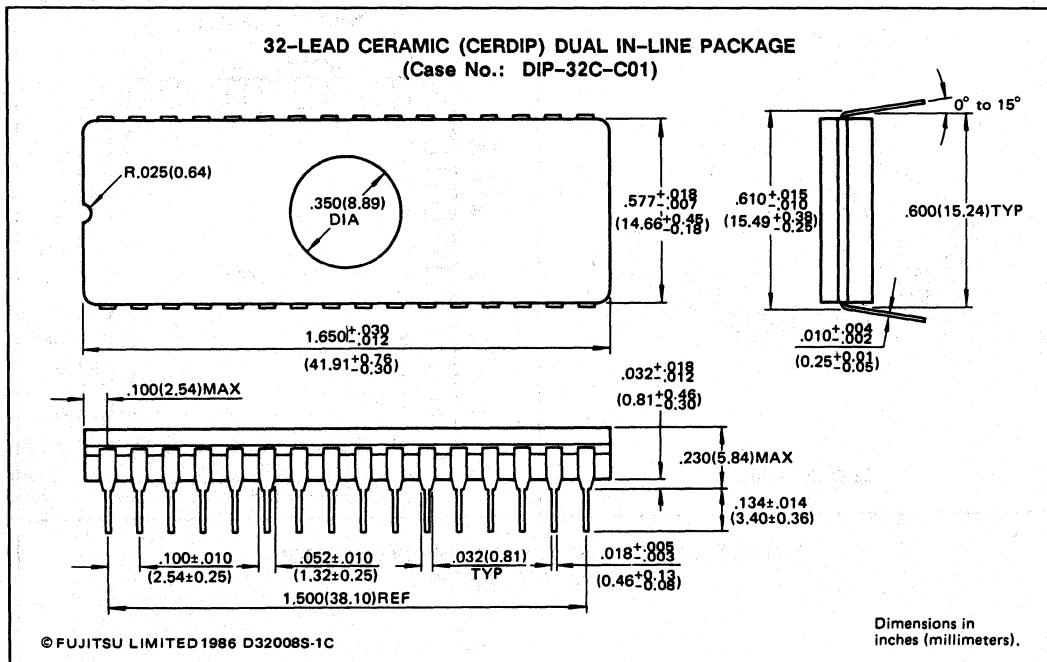
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Dimension in
inches (millimeters)



MBM27C1001-15
MBM27C1001-20
MBM27C1001-25

PACKAGE DIMENSIONS



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